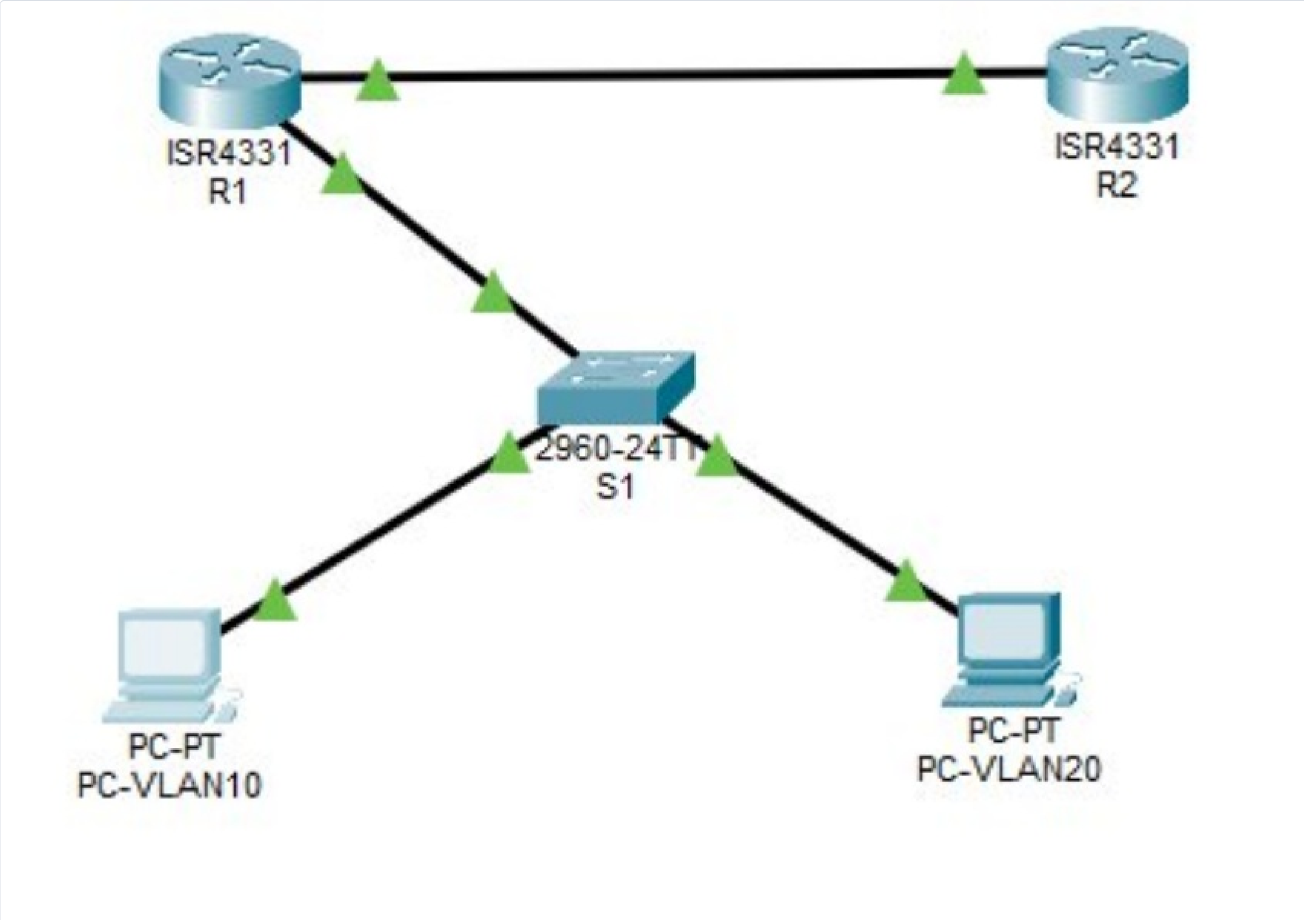
IPv6 Inter-VLAN Routing & OSPFv3

Focus: architectural deployment, incident analysis (troubleshooting), and Layer 3 hardening.



1. Lab Context & Technology Objectives

This hands-on lab simulates migrating and securing an enterprise network segment to a pure IPv6 architecture. The goal is to validate Inter-VLAN routing via the Router-on-a-Stick (Dot1Q) approach and dynamically propagate the network topology using the **OSPFv3** (Open Shortest Path First version 3) link-state routing protocol.

As a Security Consultant / SOC Analyst, this lab is an opportunity to master transition-flow auditing, analyze how switches and routers behave when ARP is replaced by **NDP** (Neighbor Discovery Protocol), and document Cisco IOS syntax implementation errors.

2. Physical Topology & Addressing Plan

The architecture consists of two Cisco ISR 4331 routers, a Catalyst 2960 switch, and two workstations simulating isolated departments.

- **Inter-router link (WAN):** R1 (Gi0/0/1) <--> R2 (Gi0/0/1)
- **Local trunk link:** R1 (Gi0/0/0) <--> S1 (Gi0/1)
- **Client hosts:** PC-VLAN10 on Fa0/10, PC-VLAN20 on Fa0/20

Device / Interface	VLAN	IPv6 Prefix / Length	IPv6 Address	Default Gateway
R1 – Gi0/0/0.10	10 (HR)	`2001:DB8:1:10::/64`	`2001:DB8:1:10::254`	N/A
R1 – Gi0/0/0.20	20 (PROD)	`2001:DB8:1:20::/64`	`2001:DB8:1:20::254`	N/A
R1 – Gi0/0/1	N/A	`2001:DB8:1:12::/64`	`2001:DB8:1:12::1`	N/A
R2 – Gi0/0/1	N/A	`2001:DB8:1:12::/64`	`2001:DB8:1:12::2`	N/A
PC-VLAN10	10	`2001:DB8:1:10::/64`	`2001:DB8:1:10::10`	`2001:DB8:1:10::254`
PC-VLAN20	20	`2001:DB8:1:20::/64`	`2001:DB8:1:20::10`	`2001:DB8:1:20::254`

3. Step-by-Step Configuration

3.1 Layer 2 Configuration (Switch S1)

Create the VLAN database, assign access ports, and bring up the 802.1Q trunk link to router R1.

```
Switch> enable
Switch# configure terminal
Switch(config)# hostname S1

! --- Create isolated segments ---
S1(config)# vlan 10
S1(config-vlan)# name RH
S1(config-vlan)# vlan 20
S1(config-vlan)# name PROD
S1(config-vlan)# exit

! --- Assign access interfaces ---
S1(config)# interface fastEthernet 0/10
S1(config-if)# switchport mode access
S1(config-if)# switchport access vlan 10
S1(config-if)# exit

S1(config)# interface fastEthernet 0/20
S1(config-if)# switchport mode access
S1(config-if)# switchport access vlan 20
S1(config-if)# exit

! --- Enable the trunk link to edge router R1 ---
S1(config)# interface gigabitEthernet 0/1
S1(config-if)# switchport mode trunk
S1(config-if)# no shutdown
```

3.2 Edge Router Configuration (R1)

The order of operations matters here: IPv6 routing must be enabled globally first, then the sub-interfaces configured with the correct Dot1Q encapsulation, **before** establishing OSPFv3 adjacency.

```
Router> enable
Router# configure terminal
Router(config)# hostname R1

! --- CRITICAL STEP: enable global IPv6 switching ---
R1(config)# ipv6 unicast-routing

! --- Instantiate the OSPFv3 process ---
R1(config)# ipv6 router ospf 1
R1(config-rtr)# router-id 1.1.1.1
R1(config-rtr)# exit

! --- Router-on-a-Stick configuration (LAN-facing link) ---
R1(config)# interface gigabitEthernet 0/0/0
R1(config-if)# no shutdown
R1(config-if)# exit

! VLAN 10 sub-interface (HR)
R1(config)# interface gigabitEthernet 0/0/0.10
R1(config-subif)# encapsulation dot1Q 10
R1(config-subif)# ipv6 address 2001:DB8:1:10::254/64
R1(config-subif)# ipv6 ospf 1 area 0
R1(config-subif)# exit

! VLAN 20 sub-interface (PROD)
R1(config)# interface gigabitEthernet 0/0/0.20
R1(config-subif)# encapsulation dot1Q 20
R1(config-subif)# ipv6 address 2001:DB8:1:20::254/64
R1(config-subif)# ipv6 ospf 1 area 0
R1(config-subif)# exit

! --- WAN link to R2 ---
R1(config)# interface gigabitEthernet 0/0/1
R1(config-if)# ipv6 address 2001:DB8:1:12::1/64
R1(config-if)# ipv6 ospf 1 area 0
R1(config-if)# no shutdown
```

3.3 Core Router Configuration (R2)

R2 faces R1. Its routing process must be configured symmetrically to negotiate the adjacency and ingest the full topology database via backbone area 0.

```
Router> enable
Router# configure terminal
Router(config)# hostname R2

! --- Enable IPv6 processing ---
R2(config)# ipv6 unicast-routing

! --- Start the OSPFv3 process ---
R2(config)# ipv6 router ospf 1
R2(config-rtr)# router-id 2.2.2.2
R2(config-rtr)# exit

! --- Configure the WAN interconnection interface ---
R2(config)# interface gigabitEthernet 0/0/1
R2(config-if)# ipv6 address 2001:DB8:1:12::2/64
R2(config-if)# ipv6 ospf 1 area 0
R2(config-if)# no shutdown
```

4. Incident Log & Resolutions

The real value of a deployment lies in identifying and correcting the infrastructure's deviant behavior. Below is the map of incidents resolved during this lab.

Incident #1 — Global OSPFv3 instantiation blocked

- **Symptom:** `% OSPFv3: IPv6 routing not enabled`
- **Root cause:** by default, Cisco IOS firmware operates exclusively at Layer 2 / IPv4 Layer 3. Running a next-generation dynamic routing protocol requires the global IPv6 address-switching daemon to be started first.
- **Fix:** run `ipv6 unicast-routing` in global configuration mode before declaring any protocol.

Incident #2 — Automatic process-ID allocation failure

- **Symptom:** `%OSPFv3-4-NORTRID: OSPFv3 process 1 could not pick a router-id`
- **Root cause:** the OSPFv3 standard uses a 32-bit identification field (same format as an IPv4 address) to map the router in the LSDB. On a network configured purely for IPv6, the router has no IPv4 address to clone from, freezing protocol startup.
- **Fix:** manually assign a unique identifier at the router prompt: `router-id 1.1.1.1` (R1) and `router-id 2.2.2.2` (R2).

Incident #3 — Interface OSPF command rejected

- **Symptom:** `% OSPFv3: No IPv6 enabled on this interface`
- **Root cause:** command sequencing error. The operator attempted to add the physical or logical sub-interface to the OSPF area before assigning it a global unicast address block. Cisco does not allow dynamic routing to listen on an interface with no active IP stack.
- **Fix:** reverse the order — declare the IPv6 address with `ipv6 address [PREFIX]` first, then immediately call the area directive `ipv6 ospf 1 area 0`.

Incident #4 — Persistent physical shutdown state on the local trunk

- **Root cause:** a classic hardware-layer activation oversight. The virtual sub-interfaces (`0/0/0.10` and `0/0/0.20`) were fully configured, but the parent interface (`GigabitEthernet0/0/0`) remained in Cisco's default software shutdown state.
- **Fix:** apply `no shutdown` directly on the parent interface Gi0/0/0. This immediately brings the encapsulated virtual links to the operational Up/Up state.

5. Audit & Operational Validation Commands

To prove the infrastructure works end-to-end — for a supervision team or a technical portfolio — the following verification commands should return specific results:

- **Routing adjacency check:** on R2, `show ipv6 ospf neighbor` must show router R1 (`1.1.1.1`) with a state of `FULL`.
- **Global routing table check:** `show ipv6 route` on R2 must list both remote subnets (VLAN 10 and VLAN 20) flagged `O` (OSPF), proving R1 correctly injects the segments.
- **End-to-end data plane validation:** a ping from the PC-VLAN10 terminal to PC-VLAN20's address (`ping 2001:DB8:1:20::10`) must succeed 100% of the time. The first packet goes through NDP for Layer 2 address resolution before the communication is confirmed.